

Rescue to Adoption: Executive Analytics Dashboard

End-to-End SQL & Power BI Business Intelligence Project

Cover Page

Project Name

Rescue to Adoption: Executive Analytics Dashboard

Project Type

End-to-End Business Intelligence & Data Analytics Project

Domain

Animal Welfare & Rescue Operations

Technology Stack

- MySQL
- SQL
- Power BI Desktop
- Common Table Expressions (CTEs)
- SQL Window Functions
- Data Cleaning
- Business Intelligence
- Git & GitHub

Project Status

Completed

Document Version

1.0

Prepared By

Portfolio Project Documentation

Purpose of this Document

This document serves as the master technical and business documentation for the **Rescue to Adoption: Executive Analytics Dashboard** project. It is designed to preserve the complete implementation, analytical thinking, business logic, and technical workflow of the project in a single reference document.

The primary objective of this documentation is to ensure that the project can be understood, explained, maintained, enhanced, or recreated without reopening the original project files. It captures both the business perspective and the technical implementation, making it suitable for interview preparation, portfolio presentation, future enhancements, and long-term project maintenance.

Rather than functioning as a traditional academic report, this document follows an industry-oriented documentation approach by explaining the rationale behind technical decisions, business requirements, analytical methodologies, implementation strategies, and the overall solution architecture.

The document is intended to serve as the single source of truth for the project and can be used to:

- Explain the project during technical and managerial interviews.
- Understand the end-to-end implementation workflow.
- Demonstrate business intelligence and analytical capabilities.
- Support future project enhancements and maintenance.
- Recreate the analytical solution if required.
- Generate resume content, portfolio summaries, GitHub documentation, and presentation material.

Revision History

Version Date		Description
1.0	Initial Release	Complete documentation of the Rescue to Adoption Executive Analytics Dashboard project.

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Project Identity

Project Name

Rescue to Adoption: Executive Analytics Dashboard

Project Category

Business Intelligence | SQL Analytics | Executive Dashboard | Data Analytics Portfolio

Domain

Animal Welfare & Rescue Operations

Business Function

Operational Analytics, Adoption Performance Analysis, Medical Resource Optimization, Executive Reporting, Decision Support

Technology Stack

Database

- MySQL

Data Processing

- SQL
- Common Table Expressions (CTEs)
- Aggregate Functions
- Window Functions
- Multi-table Joins
- Data Cleaning

Business Intelligence

- Power BI Desktop

Data Visualization

- KPI Cards
- Matrix Scorecards
- Interactive Slicers
- Ranked Bar Charts
- Pareto Analysis
- Scatter Plot Analysis

Version Control

- Git
 - GitHub
-

Project Objective

The objective of this project is to develop an executive-level business intelligence solution that enables an animal welfare organization to monitor adoption performance, evaluate medical

expenditure, analyze operational efficiency, and support strategic decision-making through interactive dashboards and analytical SQL reporting.

The project integrates SQL-based analytical processing with Power BI dashboard development to transform operational data into meaningful business insights. Multiple analytical layers were developed to answer key management questions related to animal adoption, healthcare costs, breed performance, city-wise operations, and resource allocation.

The final solution emphasizes business storytelling rather than simple visualization, allowing management to identify operational bottlenecks, monitor performance indicators, evaluate healthcare investments, and prioritize improvement initiatives using data-driven insights.

Project Architecture

The project follows a structured analytical workflow in which each stage builds upon the previous stage. Instead of directly developing dashboards from raw data, the project separates data preparation, business analysis, reporting datasets, and visualization into dedicated modules.

The overall implementation consists of the following stages:

Project Initialization

Collection of project requirements, business objectives, datasets, and analytical scope.



Data Preparation

Database creation, data import, validation, cleaning, and quality assessment.



SQL Business Analytics

Development of analytical SQL queries focusing on adoption analysis, medical expenditure, operational efficiency, breed performance, and executive KPIs.



Dashboard Data Layer

Preparation of optimized SQL outputs specifically designed for Power BI reporting.



Power BI Development

Creation of interactive executive dashboards using KPI cards, scorecards, Pareto analysis, scatter plots, ranked visualizations, and interactive slicers.



Portfolio & Documentation

GitHub repository creation, dashboard documentation, README preparation, screenshots, and portfolio assets.

Executive Summary

The **Rescue to Adoption: Executive Analytics Dashboard** is an end-to-end Business Intelligence project developed to transform operational animal welfare data into actionable management insights using SQL and Power BI.

Animal welfare organizations generate large volumes of operational data covering rescued animals, adoptions, medical treatments, healthcare expenditure, and shelter operations. However, raw operational data alone provides limited value unless it is systematically transformed into meaningful information that supports strategic planning and informed decision-making.

This project addresses that challenge by designing a complete analytics pipeline that begins with structured SQL analysis and culminates in an interactive executive dashboard. The analytical solution focuses on measuring organizational performance through key business metrics such as adoption rates, medical resource utilization, breed-level healthcare performance, operational efficiency, and city-wise welfare investments.

The implementation follows a layered architecture in which raw datasets are first validated and cleaned before analytical SQL modules are developed. These modules generate optimized reporting datasets that serve as the foundation for Power BI visualizations. By separating data preparation, business logic, and dashboard development, the project ensures maintainability, scalability, and consistency across analytical outputs.

The final dashboard enables management to:

- Monitor overall adoption performance across the organization.
- Evaluate healthcare expenditure at city and breed levels.
- Identify breeds requiring additional operational attention.
- Understand resource concentration using Pareto (80/20) analysis.
- Evaluate whether higher medical investments contribute to improved adoption outcomes.
- Support executive decision-making through interactive business intelligence reporting.

The project demonstrates practical application of SQL analytical techniques, data modeling principles, dashboard design, and business storytelling while maintaining a strong focus on executive reporting rather than descriptive visualization.

Business Background

Animal rescue and welfare organizations operate in an environment where financial resources, medical facilities, shelter capacity, and volunteer efforts must be allocated efficiently to maximize positive outcomes for rescued animals.

Every rescued animal progresses through multiple operational stages, including registration, medical evaluation, treatment, rehabilitation, and eventually adoption. Throughout this lifecycle, organizations collect valuable operational data related to healthcare costs, medical visits, adoption records, breed characteristics, and shelter locations.

Despite having access to substantial operational data, many organizations rely on manual reporting methods that make it difficult to identify performance trends, evaluate resource utilization, or prioritize operational improvements. Without centralized analytical reporting, management often lacks visibility into critical business questions such as:

- Which cities require higher medical investments?
- Which breeds generate the highest healthcare costs?
- Are healthcare resources being utilized efficiently?
- Which operational areas require immediate management attention?
- Are adoption initiatives producing expected outcomes?

Business Intelligence provides a structured approach to answering these questions by transforming operational data into meaningful business information. Rather than focusing solely on historical reporting, analytical dashboards enable organizations to identify trends, monitor performance, and make proactive decisions based on reliable data.

The project adopts this business perspective by treating the animal welfare organization as a data-driven enterprise where operational efficiency, healthcare investments, and adoption performance can all be measured using objective analytical metrics.

Problem Statement

The organization maintains multiple operational datasets covering rescued animals, adoption records, healthcare activities, and medical expenditures. Although these datasets contain valuable business information, they exist as isolated operational records without integrated analytical reporting.

As a result, management faces several challenges:

- Lack of centralized executive reporting for monitoring organizational performance.
- Limited visibility into healthcare expenditure across cities and animal breeds.
- Difficulty identifying breeds requiring operational intervention.
- Absence of standardized KPIs for measuring adoption performance.
- Inability to evaluate whether healthcare investments contribute to improved adoption outcomes.
- Limited analytical capability for prioritizing operational improvements based on business impact.

Traditional tabular reports make it difficult for decision-makers to identify trends, compare operational performance, or recognize resource concentration across different dimensions of the organization.

To address these limitations, an integrated analytical solution is required that combines structured SQL analysis with interactive business intelligence reporting, enabling management to monitor organizational performance through meaningful KPIs and executive dashboards.

System / Solution Overview

The project implements a complete Business Intelligence workflow that transforms operational data into executive-level analytical reporting.

The implementation follows a structured multi-stage architecture where each analytical layer builds upon validated outputs from the previous stage.

Stage 1 – Data Acquisition

Operational datasets representing animal records, adoption information, medical treatments, and related organizational data were collected and imported into a MySQL database.

The imported datasets formed the foundational data layer for all subsequent analytical processing.

Stage 2 – Data Validation and Cleaning

Before performing analytical calculations, the datasets were systematically reviewed to identify data quality issues.

Validation activities included:

- Verification of missing values.
- Duplicate record detection.
- Data type validation.
- Consistency checks across related tables.
- Standardization of categorical values.

This stage ensured that downstream analytical outputs were generated using reliable and consistent data.

Stage 3 – SQL Business Analytics

Multiple SQL analytical modules were developed to answer specific business questions related to organizational performance.

The implementation incorporated:

- Aggregate functions for KPI generation.
- Multi-table joins for integrated reporting.
- Common Table Expressions (CTEs) for modular query design.
- Window Functions for cumulative analysis and ranking.
- Business calculations for executive reporting metrics.

Each SQL module produced optimized reporting datasets that could be consumed directly by Power BI.

Stage 4 – Dashboard Data Layer

Instead of connecting Power BI directly to complex SQL logic, dedicated reporting datasets were exported from SQL.

This intermediate reporting layer simplified dashboard development while ensuring that all business calculations remained centralized within SQL.

The reporting datasets represented finalized analytical outputs optimized for visualization and executive reporting.

Stage 5 – Executive Dashboard Development

Power BI was used to transform analytical datasets into an interactive executive dashboard.

The dashboard combines KPI scorecards, ranked visualizations, analytical matrices, Pareto analysis, scatter plot analysis, and interactive slicers to support business exploration and executive decision-making.

Rather than presenting raw operational data, the dashboard emphasizes business interpretation by enabling management to evaluate organizational performance from multiple perspectives, including adoption performance, healthcare investment, operational efficiency, and resource concentration.

Solution Benefits

The implemented solution delivers several business advantages:

- Centralized executive reporting.
- Standardized KPI monitoring.
- Improved visibility into operational performance.
- Data-driven resource allocation.
- Better understanding of healthcare expenditure patterns.
- Enhanced analytical capability for strategic decision-making.
- Interactive reporting through self-service business intelligence.

By integrating SQL analytics with Power BI visualization, the project demonstrates how operational data can be transformed into an executive decision-support system that enables more informed management of animal welfare operations.

Dataset / Data Source Documentation

Dataset Overview

The project is built using a relational dataset representing the operational activities of an animal welfare organization. The data captures the complete lifecycle of rescued animals, including registration, adoption activities, medical treatment records, shelter operations, and geographical information required for business reporting.

Instead of relying on a single flat dataset, the project utilizes multiple related tables to simulate a real-world operational database. This relational structure enables complex business analysis through SQL joins while maintaining data integrity and reducing redundancy.

The datasets collectively support the generation of executive KPIs, operational analytics, healthcare expenditure analysis, adoption performance reporting, and interactive business intelligence dashboards.

Data Source

The project uses structured CSV datasets imported into a MySQL relational database. Each dataset represents a specific business entity within the animal rescue ecosystem.

The imported tables form the primary analytical data source for all SQL queries and Power BI dashboards.

Primary Database Tables

Animals

The Animals table serves as the master entity for the entire project. Each record represents a rescued animal and contains descriptive information used throughout analytical reporting.

The table provides:

- Animal identification
- Animal type
- Breed information
- Demographic attributes
- Reference key for operational records

This table acts as the central hub connecting adoption records and medical history.

Adoptions

The Adoptions table stores information about successful animal adoptions.

It is primarily used to measure:

- Adoption volume
- Adoption rate
- Repeat adopters
- Adoption performance by breed
- Adoption performance by animal type

The table establishes the relationship between rescued animals and adopters.

Medical Records

The Medical Records table captures healthcare activities performed on rescued animals.

This table provides the foundation for medical expenditure analysis through information such as:

- Medical visits
- Treatment records
- Treatment costs
- Healthcare utilization

Medical expenditure KPIs generated in SQL are primarily derived from this dataset.

Supporting Tables

Additional supporting tables provide business context required for operational reporting.

These datasets enable:

- City-wise analysis
- Operational segmentation
- Executive reporting
- Relationship-based SQL analysis

The relational design ensures that analytical calculations can be performed efficiently using SQL joins rather than duplicated data.

Data Model / Database Architecture

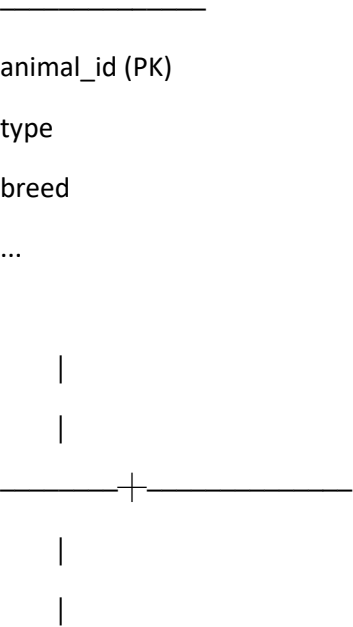
The project follows a normalized relational database structure in which operational information is distributed across multiple related tables.

Rather than storing all information in a single dataset, individual business entities are maintained independently and connected through primary and foreign key relationships.

This design improves data consistency, minimizes redundancy, and supports scalable analytical reporting.

The simplified relationship model is illustrated below.

Animals



Adoptions	Medical_Records
adoption_id (PK)	record_id (PK)
animal_id (FK)	animal_id (FK)
adopter_id	treatment_cost
...	...

The **Animals** table functions as the central fact reference for the project.

Both **Adoptions** and **Medical Records** reference the Animals table using **animal_id**, enabling integrated business analysis across operational domains.

This relational architecture allows SQL queries to combine healthcare, adoption, and animal information into a unified analytical model.

Data Quality Assessment

Before beginning analytical development, the imported datasets underwent a structured data quality assessment to ensure reliability and consistency.

The objective of this phase was to identify issues that could affect KPI calculations, SQL reporting, or dashboard accuracy.

The assessment focused on the following quality dimensions:

Missing Values

Columns were reviewed for missing or incomplete values that could influence business calculations.

Special attention was given to:

- Animal identifiers
- Breed information
- Adoption records
- Medical costs
- Foreign key columns

Missing values were handled according to business relevance before analytical reporting was performed.

Duplicate Records

Duplicate detection was performed to verify entity uniqueness and prevent inflated business metrics.

Validation included:

- Primary key verification
- Composite duplicate identification
- Business entity validation

Ensuring uniqueness was particularly important for adoption counts, medical visit calculations, and executive KPIs.

Data Type Validation

All imported columns were reviewed to ensure appropriate SQL data types.

Validation included:

- Numeric fields
- Date fields
- Text attributes
- Identifier columns
- Cost-related measures

Correct data types were necessary for aggregation, filtering, sorting, and dashboard visualization.

Referential Integrity

Relationships between operational tables were validated to confirm that foreign key values correctly referenced existing animal records.

Relationship verification ensured that SQL joins produced accurate analytical outputs without orphaned records.

Business Rule Validation

Domain-specific validation was performed to identify inconsistencies that could affect reporting quality.

Examples included:

- Invalid categorical values.
- Unknown breed classifications.
- Inconsistent operational records.
- Missing business identifiers.

These checks ensured that executive KPIs reflected meaningful business information rather than underlying data inconsistencies.

Data Cleaning & Transformation Strategy

Following data profiling, the datasets were cleaned and standardized prior to analytical processing.

The objective of this stage was not only to improve data quality but also to prepare the data for efficient SQL analysis and dashboard reporting.

The cleaning strategy focused on preserving business meaning while ensuring analytical consistency.

Standardization

Categorical values were standardized to eliminate inconsistencies caused by formatting variations.

This ensured that grouping operations, filtering, and dashboard slicers produced accurate business summaries.

Handling Missing Data

Missing values were reviewed individually to determine their impact on business reporting.

Where appropriate, incomplete records were either corrected, excluded from specific analyses, or retained when they represented meaningful business scenarios.

For example, unidentified breeds were recognized as data quality concerns and treated separately during analytical reporting rather than allowing them to distort business insights.

Relationship Validation

Before developing analytical SQL queries, relationships between operational tables were verified to ensure that records could be joined reliably.

Relationship validation prevented inaccurate KPI calculations resulting from missing or mismatched identifiers.

Analytical Dataset Preparation

Rather than connecting Power BI directly to raw operational tables, dedicated SQL queries were developed to produce reporting-ready datasets.

This approach centralized business calculations within SQL while keeping Power BI focused on visualization and executive reporting.

The reporting datasets served as the final analytical layer consumed by the dashboard, improving both performance and maintainability.

Transformation Outcome

Upon completion of the cleaning and transformation phase, the datasets were suitable for:

- Executive KPI generation
- Adoption performance analysis
- Medical expenditure reporting
- Operational efficiency analysis
- Pareto cost analysis
- Interactive dashboard development
- Business decision support

The resulting analytical data layer formed the foundation for all subsequent SQL reporting modules and Power BI visualizations.

Feature Engineering / Business Logic

Unlike machine learning projects where feature engineering focuses on creating predictive variables, this Business Intelligence project emphasizes business-driven feature engineering. New analytical fields and calculated metrics were developed within SQL to transform operational data into meaningful business information for reporting and executive decision-making.

These derived metrics represent business concepts rather than raw transactional values, allowing management to evaluate organizational performance using standardized indicators.

The primary business logic implemented throughout the project is described below.

Executive KPI Generation

Executive KPIs were developed to provide management with an immediate overview of organizational performance.

Rather than displaying raw operational records, SQL aggregation techniques were used to summarize business performance into measurable indicators.

The KPI layer includes metrics such as:

- Total Animals
- Adopted Animals
- Unadopted Animals
- Adoption Rate
- Medical Visits
- Total Medical Cost
- Average Medical Cost
- Repeat Adopters
- Critical Breeds

These KPIs form the foundation of executive reporting and provide high-level visibility into operational performance.

Adoption Performance Logic

Adoption performance is one of the primary success indicators for an animal welfare organization.

Business logic was implemented to calculate adoption-related metrics by comparing the number of successfully adopted animals against the total rescued animal population.

The resulting adoption metrics allow management to:

- Monitor organizational effectiveness.
- Compare adoption performance across different animal groups.
- Identify areas requiring operational intervention.

Medical Cost Analysis

Medical expenditure represents one of the largest operational costs within animal welfare organizations.

Business logic was implemented to aggregate treatment costs recorded in medical records and generate multiple analytical measures, including:

- Total medical expenditure
- Average treatment cost
- Medical visit counts
- Breed-wise healthcare investment
- City-wise healthcare expenditure

These metrics support financial monitoring and resource allocation decisions.

Breed Performance Evaluation

Breed-level analysis was introduced to evaluate healthcare investment alongside adoption performance.

Instead of focusing solely on adoption counts, the project measures overall breed performance using multiple business indicators, including:

- Adoption rate
- Medical visits
- Total healthcare expenditure
- Average healthcare cost

This multidimensional evaluation enables management to understand which breeds require greater operational attention while balancing both welfare outcomes and financial investment.

Risk Classification Logic

To simplify executive interpretation, breeds were categorized into performance groups using adoption rate thresholds.

Each breed is automatically assigned to one of the following categories:

- Excellent
- Good
- Needs Improvement
- Critical

The classification provides an intuitive method for prioritizing operational actions without requiring managers to manually interpret numerical adoption rates.

Risk categorization also enables conditional formatting within the Power BI dashboard, allowing lower-performing breeds to be identified quickly.

Pareto Analysis (80/20 Rule)

A Pareto analysis was implemented to identify the breeds contributing the largest proportion of medical expenditure.

Using SQL Window Functions, cumulative contribution percentages were calculated to determine the concentration of healthcare costs across breeds.

Each breed was classified as either:

- Top 80% Cost Driver
- Low Cost Contributor

This business logic enables management to focus improvement initiatives on the relatively small group of breeds responsible for the majority of healthcare spending.

The implementation demonstrates how statistical business principles can be applied using SQL analytical functions.

Operational Efficiency Metrics

Operational efficiency was evaluated by combining information from adoption records and medical expenditure.

Rather than analyzing healthcare spending independently, the project compares medical investment with adoption outcomes to determine whether organizational resources are being utilized effectively.

This approach supports strategic decision-making by identifying situations where higher operational investment may not correspond to improved adoption performance.

Dashboard Reporting Layer

Instead of performing business calculations directly within Power BI, reporting datasets were prepared using SQL.

This design centralizes business logic within the database layer while keeping the visualization layer focused on reporting and interaction.

The reporting datasets were structured specifically for dashboard consumption, improving maintainability and reducing duplication of calculations.

Technical Implementation

Development Environment

The project was implemented using a combination of relational database technologies and business intelligence tools.

Database Platform

- MySQL

The database served as the central analytical engine responsible for data storage, cleaning, transformation, and business calculations.

Business Intelligence Platform

- Microsoft Power BI Desktop

Power BI was used to create interactive executive dashboards by consuming reporting-ready datasets generated through SQL.

Version Control

- Git
- GitHub

Version control was used to maintain project history, organize documentation, and publish the completed portfolio project.

SQL Implementation Strategy

SQL forms the analytical backbone of the project.

Rather than using isolated queries, the implementation follows a modular approach where individual SQL scripts address specific business questions.

Major SQL techniques utilized throughout the project include:

- Multi-table joins
- Aggregate functions
- Common Table Expressions (CTEs)
- Window Functions
- GROUP BY
- HAVING
- CASE expressions
- Ranking calculations
- Percentage calculations
- Business classifications

This modular design improves readability while allowing individual analytical components to be maintained independently.

Common Table Expressions (CTEs)

CTEs were used extensively to improve query readability and organize complex analytical calculations into logical stages.

Instead of embedding multiple nested subqueries, intermediate analytical results were generated using CTEs before producing final reporting outputs.

This approach enhances maintainability and simplifies future modifications.

Window Functions

Window Functions were implemented to support advanced analytical reporting.

Key applications include:

- Running cumulative percentages
- Pareto analysis
- Ranking calculations
- Percentage contribution analysis

These functions enabled sophisticated business analytics without requiring procedural programming.

Aggregate Analysis

Aggregate functions were used throughout the project to summarize operational data into executive metrics.

Examples include:

- Total counts
- Distinct counts
- Average calculations
- Cost aggregation
- Percentage calculations

Aggregation transformed detailed transactional records into concise business information suitable for executive reporting.

Conditional Business Logic

CASE expressions were implemented to convert numerical business metrics into meaningful business classifications.

Examples include:

- Risk categories
- Performance groups

- Pareto classifications

This logic improves dashboard readability by presenting business-friendly categories instead of raw numerical values.

Dashboard Dataset Preparation

Dedicated SQL queries were developed to generate reporting-ready datasets optimized for Power BI.

Each reporting dataset addresses a specific analytical requirement, reducing unnecessary complexity within the visualization layer.

Separating analytical computation from dashboard development also improves performance and ensures consistent business calculations across reports.

Execution Workflow

The project followed a structured end-to-end analytics workflow designed to mirror a real-world Business Intelligence implementation.

Step 1 – Business Understanding

Project objectives and business questions were identified to define the analytical scope.

Key reporting requirements focused on adoption performance, healthcare expenditure, operational efficiency, and executive KPI reporting.

Step 2 – Database Preparation

Operational datasets were imported into MySQL and organized into a relational database structure.

Relationships between business entities were validated before analytical development began.

Step 3 – Data Quality Assessment

The imported datasets underwent profiling to identify missing values, duplicate records, inconsistent categories, and relationship issues.

Necessary cleaning activities were completed before business analysis.

Step 4 – SQL Business Analytics

Business questions were translated into modular SQL queries.

Each analytical module generated reporting datasets focused on a specific operational objective, including adoption analysis, healthcare expenditure, executive KPIs, and resource optimization.

Step 5 – Dashboard Dataset Creation

Final SQL outputs were exported as reporting-ready datasets.

These datasets formed the dedicated data layer for Power BI development, ensuring that business logic remained centralized within SQL.

Step 6 – Executive Dashboard Development

Interactive dashboards were designed in Power BI using KPI cards, analytical matrices, ranked visualizations, Pareto analysis, scatter analysis, and slicers.

The dashboard emphasizes executive decision support rather than operational reporting.

Step 7 – Portfolio Publication

The completed project was organized into a structured repository containing SQL scripts, dashboard files, documentation, screenshots, and supporting assets.

Version control through Git and GitHub ensures maintainability while providing a professional portfolio artifact suitable for technical interviews and future reference.

Analytical Thinking / Design Decisions

The project was designed with the objective of creating a business intelligence solution rather than a collection of independent SQL queries or dashboard visualizations. Every implementation decision was guided by business usability, maintainability, scalability, and executive reporting requirements.

The following design principles were adopted throughout the project.

Business-First Approach

Instead of beginning with dashboard development, the project started by identifying the business questions that management would expect the reporting solution to answer.

This ensured that every SQL query, calculated metric, and dashboard visualization served a clear business purpose.

The project focused on answering questions such as:

- How effectively are rescued animals being adopted?
- Which breeds require greater operational attention?
- Where are healthcare resources being invested?
- Are healthcare investments producing meaningful adoption outcomes?
- Which operational areas require management intervention?

Designing the solution around business questions rather than charts resulted in a more practical and decision-oriented analytical system.

SQL as the Primary Analytical Engine

Business calculations were intentionally implemented within SQL instead of Power BI wherever possible.

This architectural decision provides several advantages:

- Centralized business logic
- Consistent calculations across reports
- Easier debugging
- Improved dashboard performance
- Reduced dependency on visualization-specific calculations

Power BI was therefore used primarily as the presentation layer rather than the computational layer.

Modular SQL Design

Instead of developing one large SQL script, analytical logic was divided into multiple modules based on business objectives.

Each SQL module focuses on a single analytical problem, making the implementation easier to understand, maintain, and extend.

Examples include:

- Executive KPI generation
- Adoption analytics
- Breed performance analysis
- Medical expenditure analysis
- Operational efficiency analysis
- Pareto cost analysis

This modular structure closely resembles enterprise reporting environments where reporting logic is separated into reusable analytical components.

Executive Dashboard Philosophy

The dashboard was designed for senior management rather than operational users.

Consequently, the focus remained on:

- High-level KPIs
- Interactive filtering

- Strategic insights
- Performance monitoring
- Resource allocation
- Decision support

Charts were selected based on their ability to communicate business value rather than visual complexity.

Use of Advanced SQL Features

The project intentionally incorporates SQL concepts commonly expected during data analyst interviews.

These include:

- Common Table Expressions (CTEs)
- Window Functions
- Aggregate Functions
- Conditional Logic
- Ranking
- Percentage Calculations
- Multi-table Joins

Including these techniques demonstrates practical SQL competency while solving genuine business problems.

Dashboard Simplicity

Rather than filling the dashboard with numerous visualizations, emphasis was placed on selecting a smaller number of meaningful analytical views.

Each dashboard page was designed to answer a distinct set of business questions without overwhelming the user.

This minimalist approach improves readability while allowing management to focus on the most important performance indicators.

Business Storytelling

The project follows a storytelling approach where each analytical layer builds logically upon the previous one.

The reporting flow progresses from organizational performance to detailed operational analysis.

Management first receives an overview of organizational health before drilling into specific operational drivers such as breed performance, healthcare expenditure, and resource utilization.

This structured narrative improves the interpretability of analytical findings.

Business Questions Solved

The project addresses a range of strategic and operational questions that are relevant to animal welfare organizations.

Rather than generating descriptive reports, the analytical solution provides answers that support planning, resource allocation, and performance improvement.

The major business questions addressed include:

Adoption Performance

- How many rescued animals have been successfully adopted?
 - What percentage of rescued animals have been adopted?
 - How many animals remain unadopted?
 - Which breeds demonstrate stronger adoption performance?
-

Medical Resource Utilization

- What is the total healthcare expenditure?
 - How many medical visits have been performed?
 - What is the average treatment cost?
 - Which breeds consume the greatest healthcare resources?
-

Operational Performance

- Which cities receive the highest medical investment?
 - Which operational areas require additional management attention?
 - Which breeds are categorized as critical based on adoption performance?
-

Resource Optimization

- Which breeds contribute most significantly to healthcare expenditure?
- Does increased healthcare investment result in improved adoption outcomes?
- How concentrated are healthcare costs across breeds?
- Which operational areas offer the greatest opportunity for improvement?

Executive Reporting

- What are the organization's key operational KPIs?
- Which metrics should management monitor regularly?
- Which business areas require strategic intervention?

Dashboard Implementation

The reporting solution consists of two executive dashboard pages, each designed to address a different level of business decision-making.

The dashboards emphasize business interpretation, operational monitoring, and strategic reporting rather than displaying raw transactional data.

Dashboard 1 – Animal Welfare Overview

Business Objective

Provide senior management with a consolidated overview of organizational performance through standardized KPIs and high-level operational reporting.

This dashboard enables executives to quickly assess the current state of adoption activities, healthcare expenditure, and breed-level performance.

Major Components

- Executive KPI scorecards
- Interactive city slicer
- Interactive animal type slicer
- City-wise medical investment ranking
- Breed performance scorecard

Business Value

The dashboard enables management to:

- Monitor organizational health.
 - Compare operational performance across cities.
 - Evaluate adoption effectiveness.
 - Identify breeds requiring attention.
 - Track healthcare investment.
-

Dashboard 2 – Care Excellence & Resource Insights

Business Objective

Provide deeper analytical insight into healthcare resource allocation and operational efficiency.

This dashboard focuses on identifying the factors driving healthcare expenditure while evaluating whether organizational investments contribute to improved adoption outcomes.

Major Components

- Pareto analysis of breed-level healthcare expenditure
- Adoption efficiency versus medical investment analysis
- Executive insight summary

Business Value

The dashboard enables management to:

- Identify high-cost operational drivers.
 - Evaluate resource concentration.
 - Compare investment against adoption performance.
 - Support strategic resource allocation decisions.
 - Prioritize operational improvement initiatives.
-

Key Business Insights

Analysis of the available operational data produced several business observations that can support organizational planning and resource optimization.

Adoption Performance

The organization demonstrates measurable adoption activity across multiple animal types, allowing adoption performance to be monitored using standardized executive KPIs.

Adoption metrics provide management with a consistent method for evaluating organizational effectiveness over time.

Healthcare Resource Distribution

Medical expenditure is not distributed evenly across all breeds.

Certain breeds account for a disproportionately larger share of healthcare costs, indicating opportunities for targeted operational planning and cost management.

Resource Concentration

Pareto analysis demonstrates that a relatively small group of breeds contributes the majority of medical expenditure.

Understanding this concentration enables management to focus improvement initiatives where they are likely to produce the greatest operational impact.

Adoption Efficiency

Comparing healthcare investment with adoption performance indicates that higher expenditure alone does not necessarily correspond to stronger adoption outcomes.

This highlights the importance of evaluating operational effectiveness alongside financial investment.

Executive Decision Support

Interactive KPI monitoring and analytical reporting allow management to move beyond static reporting by exploring operational performance through multiple business dimensions.

The reporting solution provides a centralized platform for monitoring adoption trends, healthcare expenditure, and breed performance while supporting data-driven strategic decisions.

Business Recommendations

Based on the analytical findings generated throughout the project, the following recommendations can support improved operational performance and more effective resource utilization.

Optimize Healthcare Investment

Regularly review breed-level medical expenditure to ensure healthcare resources are allocated efficiently.

Breeds with consistently high treatment costs should be evaluated to identify opportunities for preventive care, operational improvements, or process optimization.

Prioritize Critical Breeds

Breeds classified within lower adoption performance categories should receive targeted operational attention.

Focused adoption campaigns, enhanced rehabilitation efforts, or specialized outreach initiatives may improve adoption outcomes.

Monitor Executive KPIs

Establish a routine review process for executive KPIs to enable continuous monitoring of organizational performance.

Regular KPI tracking supports proactive management rather than reactive decision-making.

Strengthen Data Quality

Improve operational data capture processes to reduce incomplete classifications and ensure consistent reporting.

Higher-quality data improves analytical accuracy and enhances confidence in management decisions.

Expand Analytical Reporting

Future reporting can incorporate additional operational metrics such as shelter occupancy, volunteer activity, donation trends, and post-adoption follow-up outcomes.

Expanding the analytical scope would provide management with a more comprehensive understanding of organizational performance.

KPI Documentation

The executive dashboard was designed around standardized Key Performance Indicators (KPIs) that summarize organizational performance into measurable business metrics. These KPIs provide management with an immediate understanding of adoption performance, operational efficiency, and healthcare resource utilization.

Each KPI is calculated through SQL and presented within the executive dashboard to support strategic monitoring and informed decision-making.

KPI 1 – Total Animals

Definition

Represents the total number of rescued animals registered within the organization.

Business Purpose

Provides the overall operational workload and establishes the baseline for all adoption and healthcare analyses.

Business Usage

Management uses this KPI to monitor shelter activity, estimate operational capacity, and evaluate changes in rescue volume over time.

KPI 2 – Adopted Animals

Definition

Represents the number of rescued animals that have been successfully adopted.

Business Purpose

Measures the organization's ability to rehabilitate and successfully place rescued animals into permanent homes.

Business Usage

Used to evaluate adoption program effectiveness and monitor overall organizational success.

KPI 3 – Unadopted Animals

Definition

Represents rescued animals that have not yet been adopted.

Business Purpose

Provides visibility into the remaining shelter population requiring continued care and operational support.

Business Usage

Helps management estimate future shelter capacity requirements, medical resources, and adoption campaign priorities.

KPI 4 – Adoption Rate

Definition

The percentage of rescued animals successfully adopted relative to the total rescued population.

Formula

Adoption Rate = (Adopted Animals ÷ Total Animals) × 100

Business Purpose

Acts as one of the primary organizational performance indicators.

Management Usage

Used to monitor organizational effectiveness, compare adoption performance across reporting periods, and assess the impact of adoption initiatives.

KPI 5 – Medical Visits

Definition

Represents the total number of medical treatment records associated with rescued animals.

Business Purpose

Measures healthcare workload and veterinary service utilization.

Management Usage

Supports planning for veterinary staffing, operational scheduling, and healthcare capacity.

KPI 6 – Total Medical Cost

Definition

Represents the cumulative healthcare expenditure incurred across all recorded medical treatments.

Business Purpose

Provides visibility into the organization's largest operational expense category.

Management Usage

Supports budgeting, financial planning, cost monitoring, and resource allocation decisions.

KPI 7 – Average Medical Cost

Definition

Average treatment expenditure per medical record.

Formula

Average Medical Cost = Total Medical Cost ÷ Total Medical Visits

Business Purpose

Measures average healthcare spending and provides a standardized cost benchmark.

Management Usage

Helps identify unusual cost fluctuations and evaluate healthcare efficiency.

KPI 8 – Repeat Adopters

Definition

Represents adopters who have completed more than one successful adoption.

Business Purpose

Measures adopter engagement and long-term community participation.

Management Usage

Supports loyalty initiatives and identifies potential advocates for future adoption campaigns.

KPI 9 – Critical Breeds

Definition

Represents breeds with comparatively lower adoption performance based on predefined business thresholds.

Business Purpose

Highlights breeds requiring operational intervention and additional adoption efforts.

Management Usage

Supports targeted awareness campaigns, rehabilitation planning, and operational prioritization.

Challenges Faced

Developing the project involved both technical and analytical challenges that required iterative refinement before achieving the final reporting solution.

Data Quality Issues

The imported datasets contained inconsistencies that required validation before analytical development.

Challenges included:

- Missing categorical values.
- Unknown breed classifications.
- Validation of relationships between operational tables.
- Ensuring data consistency across related datasets.

These issues were addressed during the data cleaning phase to improve reporting accuracy.

Designing Meaningful Business Metrics

Transforming operational records into executive KPIs required careful consideration of business definitions and reporting requirements.

Rather than displaying raw transactional information, meaningful performance indicators were developed that accurately reflected organizational performance.

SQL Query Complexity

Several analytical requirements involved combining multiple datasets while performing aggregate calculations and cumulative analysis.

To maintain readability, Common Table Expressions (CTEs) and Window Functions were adopted in place of deeply nested queries.

This modular approach simplified maintenance and improved query clarity.

Pareto Analysis Implementation

Developing the Pareto analysis required multiple iterations.

Initial implementations highlighted challenges related to cumulative percentage calculations and visualization ordering within Power BI.

The solution was refined by restructuring the SQL logic, validating cumulative calculations, and ensuring the dashboard displayed breeds in descending order of medical expenditure.

This process ensured that the Pareto analysis accurately reflected the 80/20 distribution principle.

Power BI Formatting

Developing a clean executive dashboard required significant attention to formatting and consistency.

Areas requiring refinement included:

- KPI formatting.
- Display units.
- Conditional formatting.
- Slicer interactions.
- Color consistency.
- Visual alignment.
- Dashboard spacing.

The final design prioritized readability and executive presentation over visual complexity.

Business Storytelling

One of the most significant challenges was ensuring that the dashboard communicated a coherent business narrative rather than functioning as a collection of independent charts.

Each visualization was therefore selected based on the business question it answered, ensuring that the dashboard progressed logically from organizational overview to detailed resource analysis.

Lessons Learned

The project provided valuable experience across database analytics, business intelligence development, and executive reporting.

Several important lessons emerged during implementation.

Business Questions Should Drive Analysis

Effective dashboards begin with business objectives rather than visualization selection.

Defining management questions before writing SQL or designing dashboards resulted in a more focused and valuable analytical solution.

Data Quality Directly Influences Reporting Quality

Reliable dashboards depend on reliable data.

Investing time in data validation and cleaning significantly improved the accuracy of executive KPIs and business insights.

SQL Can Handle Complex Business Logic

Many business calculations commonly implemented in visualization tools can be efficiently handled within SQL.

Centralizing business logic within the database simplifies dashboard maintenance and improves consistency across reports.

Simplicity Improves Executive Reporting

Executive dashboards should emphasize clarity rather than maximizing the number of visualizations.

Selecting a limited number of meaningful analytical views improved usability and strengthened the overall business narrative.

Dashboard Design Is About Communication

An effective dashboard communicates information rather than merely displaying data.

Consistent formatting, logical layout, intuitive filtering, and concise business insights significantly enhance decision-making.

Advanced SQL Adds Business Value

The practical application of CTEs, Window Functions, conditional logic, and aggregate analysis demonstrated that advanced SQL techniques can solve genuine business problems while producing scalable analytical solutions.

End-to-End Thinking Is Essential

Developing the project from database design through SQL analysis, dashboard development, documentation, and GitHub publication reinforced the importance of viewing analytics as a complete business solution rather than isolated technical tasks.

This end-to-end perspective better reflects real-world Business Intelligence implementations and strengthens both technical competency and portfolio quality.

STAR Interview Version

Situation

An animal welfare organization maintains operational data covering rescued animals, adoptions, medical treatments, and healthcare expenditure. Although the organization collects significant amounts of operational information, decision-makers lack a centralized reporting system to monitor adoption performance, healthcare investment, operational efficiency, and resource allocation.

Reporting is primarily operational, making it difficult for management to identify trends, prioritize improvement initiatives, and evaluate organizational performance using standardized metrics.

Task

The objective was to design and develop an end-to-end Business Intelligence solution capable of transforming operational data into executive-level analytical reporting.

The solution needed to:

- Integrate multiple operational datasets.
- Perform data validation and cleaning.
- Develop business-oriented SQL analytics.
- Generate executive KPIs.
- Produce reporting-ready datasets.
- Design an interactive Power BI dashboard focused on strategic decision-making rather than operational reporting.

Action

A relational database was created in MySQL by importing and organizing multiple operational datasets.

The implementation included comprehensive data validation, duplicate detection, missing value assessment, relationship verification, and categorical standardization to improve analytical quality.

Business logic was implemented using SQL through modular analytical queries incorporating:

- Aggregate Functions
- Multi-table Joins
- Common Table Expressions (CTEs)
- Window Functions
- CASE Expressions

- Percentage Calculations

Dedicated SQL modules were developed for:

- Executive KPI generation
- Adoption performance analysis
- Breed performance evaluation
- Medical expenditure reporting
- Operational efficiency analysis
- Pareto (80/20) cost analysis

Instead of embedding calculations within Power BI, reporting datasets were generated directly through SQL to centralize business logic and improve dashboard performance.

An executive dashboard was then developed in Power BI using KPI scorecards, ranked business visualizations, analytical matrices, Pareto analysis, scatter plot analysis, and interactive slicers.

The dashboard was designed around business storytelling, enabling management to move from high-level organizational performance to detailed operational insights.

Result

The completed solution transformed operational records into an executive decision-support dashboard capable of monitoring adoption performance, healthcare expenditure, breed-level operational effectiveness, and resource concentration.

The project demonstrates practical competency in relational database design, SQL analytics, business intelligence reporting, dashboard development, and executive storytelling while following an end-to-end analytics workflow similar to real-world Business Intelligence implementations.

Resume Master Description

Rescue to Adoption: Executive Analytics Dashboard | MySQL, SQL, Power BI

Designed and developed an end-to-end Business Intelligence solution for analyzing animal welfare operations using MySQL and Power BI. Built modular SQL analytics utilizing CTEs, Window Functions, aggregate functions, and multi-table joins to generate executive KPIs, adoption performance metrics, breed-level healthcare analysis, and Pareto-based resource optimization. Developed an interactive executive dashboard featuring KPI scorecards, analytical matrices, ranked visualizations, scatter analysis, and executive insights to support data-driven operational decision-making.

ATS Keywords

Business Intelligence

- Business Intelligence

- Executive Dashboard
 - Dashboard Development
 - KPI Reporting
 - Interactive Reporting
 - Business Analytics
 - Operational Analytics
 - Data Visualization
 - Decision Support
 - Executive Reporting
-

SQL

- MySQL
 - SQL
 - Relational Database
 - Database Design
 - Data Modeling
 - SQL Joins
 - Aggregate Functions
 - Window Functions
 - Common Table Expressions
 - CTE
 - CASE Statements
 - GROUP BY
 - HAVING
 - Analytical SQL
 - Query Optimization
-

Power BI

- Power BI Desktop
- Dashboard Design
- KPI Cards

- Matrix Visualization
 - Pareto Analysis
 - Scatter Plot
 - Interactive Slicers
 - Business Storytelling
 - Report Design
-

Data Analytics

- Data Cleaning
 - Data Validation
 - Data Transformation
 - Data Profiling
 - Data Quality
 - Business Logic
 - Exploratory Analysis
 - Operational Reporting
 - Resource Optimization
 - Performance Analysis
-

Professional Skills

- Problem Solving
 - Business Analysis
 - Critical Thinking
 - Stakeholder Reporting
 - Decision Making
 - Analytical Thinking
 - Executive Communication
-

Skills Demonstrated

Database & SQL

- Relational Database Design

- SQL Query Development
 - Multi-table Joins
 - Aggregate Analysis
 - Window Functions
 - Common Table Expressions
 - Conditional Business Logic
 - Reporting Dataset Creation
-

Data Preparation

- Data Import
 - Data Cleaning
 - Data Validation
 - Duplicate Detection
 - Missing Value Assessment
 - Data Quality Evaluation
 - Relationship Validation
-

Business Intelligence

- Executive Dashboard Design
 - KPI Development
 - Interactive Reporting
 - Business Storytelling
 - Dashboard Optimization
 - Executive Reporting
-

Analytical Skills

- Operational Performance Analysis
- Adoption Analytics
- Healthcare Cost Analysis
- Resource Allocation Analysis
- Pareto (80/20) Analysis

- Performance Classification
 - Business Recommendation Development
-

Professional Practices

- Modular Development
 - Version Control (Git & GitHub)
 - Project Documentation
 - Portfolio Development
 - Business Communication
-

Technical Interview Questions

SQL

1. Why did you choose to implement business calculations in SQL rather than Power BI?
 2. Explain the purpose of the Common Table Expressions used in your analytical queries.
 3. How were Window Functions used in your Pareto analysis?
 4. What advantages do Window Functions provide compared to GROUP BY?
 5. How did you calculate adoption rate?
 6. Why did you use LEFT JOIN in certain analytical queries?
 7. How did you identify repeat adopters?
 8. Explain the logic behind your risk classification.
 9. How did you calculate cumulative contribution percentages?
 10. How would you optimize your SQL queries if the database contained millions of records?
-

Power BI

1. Why did you separate SQL processing from dashboard development?
2. How did you design your dashboard for executive users?
3. Why did you select Pareto analysis instead of another visualization?
4. How were slicers used to improve report usability?
5. How would you improve dashboard performance for larger datasets?
6. Why were only two dashboard pages developed instead of multiple pages?
7. How would you implement Row-Level Security if required?

8. What additional DAX measures would you introduce if the dashboard were expanded?
-

Business Interview Questions

1. Why is adoption rate an important organizational KPI?
 2. How can management use medical expenditure analysis for budgeting?
 3. What business value does Pareto analysis provide?
 4. Why is executive reporting different from operational reporting?
 5. How can this dashboard improve organizational decision-making?
 6. Which business recommendation would you prioritize first and why?
 7. What additional KPIs would strengthen the reporting solution?
 8. How would this solution scale across multiple shelters?
-

HR Interview Questions

1. Why did you choose this domain for your portfolio project?
 2. Which part of the project challenged you the most?
 3. What did this project teach you about Business Intelligence?
 4. If you had another month to improve the project, what would you add?
 5. Which technology did you enjoy working with the most?
 6. How did you ensure the dashboard remained business-focused?
 7. What would you do differently if rebuilding this project from scratch?
-

Cross Questions

These are follow-up questions commonly asked after discussing the project:

- Why did you perform calculations in SQL instead of DAX?
- Why did you use CTEs instead of subqueries?
- How did you validate the correctness of your SQL outputs?
- How would your solution change if the data were refreshed daily?
- What indexing strategy would you recommend for larger datasets?
- How would you handle slowly changing dimensions in this model?
- Why is Pareto analysis useful in operational analytics?
- How would you convert this solution into a live reporting system?

- What assumptions did you make while developing the business logic?
 - Which KPI do you believe provides the greatest business value, and why?
-

Future Enhancements

The current implementation provides a strong foundation for executive reporting. Future versions of the project could extend both the analytical depth and operational capabilities.

Potential enhancements include:

- Implementing a live database connection instead of static reporting datasets.
 - Developing incremental data refresh pipelines.
 - Creating historical trend analysis across months and years.
 - Adding donation and funding analytics.
 - Monitoring shelter occupancy and capacity utilization.
 - Tracking volunteer participation and operational workload.
 - Introducing predictive models for adoption probability.
 - Implementing automated alerts for critical operational KPIs.
 - Adding Row-Level Security (RLS) for role-based dashboard access.
 - Publishing the dashboard through the Power BI Service for organizational sharing.
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Portfolio Assets

The completed portfolio package includes:

- Power BI Executive Dashboard (.pbix)
- Dashboard PDF Export
- SQL Scripts
- Reporting Datasets
- Dashboard Screenshots
- GitHub Repository
- Project README
- Project Master Documentation

These assets collectively demonstrate an end-to-end Business Intelligence workflow encompassing database analytics, SQL development, dashboard design, documentation, and portfolio presentation.

References

- Microsoft Power BI Desktop Documentation
- MySQL 8.0 Reference Manual
- SQL Window Function Documentation
- SQL Common Table Expression (CTE) Documentation
- Data Visualization Best Practices for Executive Reporting
- Pareto Principle (80/20 Rule) in Business Analytics

End of Project Master Documentation